DOTTORATO INDUSTRIALE IN BIG DATA ED INTELLIGENZA ARTIFICIALE (XXXVIII CICLO)

Curricula "Big data management per la transizione digitale" Università delle Camere di Commercio Italiane "UNIVERSITAS MERCATORUM"

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TITOLO PROGETTO: RICONOSCIMENTO DI DEEPFAKE GENERATI TRAMITE AI

[BOZZA] DEEPFAKE VIDEO DETECTION USING THE COMBINATION OF 3 DATASETS

The process includes the data collection, feature extraction, model architecture and training the inference.

Data Collection:

1. 2115 - FaceForensics ++
2. 2347 - DeepFake Detection challenge
3. 1989 - Celeb dataset

Preprocessing

It Includes face detection and cropping in videos and selecting a threshold for the number of frames based on computational limits. Here we used 20 frames sequence to train the LSTM model.

Data Split:

- Train 60%: 3856
- Valid 20%: 1285
- Test 20%: 1286

Deepfake detection system that utilises a combination of Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) with Long Short-Term Memory (LSTM) units to identify manipulated videos. Here's how LSTM works in this context:

Model Architecture:

Feature Extraction: utilises a ResNext Convolution neural network to extract frame-level features.

Sequence Analysis: employs a Long Short-Term Memory (LSTM)-based Recurrent Neural Network to analyse the sequences of frames for manipulation signs.

Dataset Balance: trains on a balanced dataset with an equal number of real and fake videos to prevent model bias.

Classification: the LSTM network, trained on a balanced dataset of real and fake videos, classifies a video as deepfake or real based on the learned temporal patterns.

The LSTM's ability to handle sequential data makes it well-suited for analysing the frames of a video over time, which is crucial for detecting deepfakes that may have inconsistencies across frames. The article also mentions the creation of a processed dataset containing only face-cropped videos, which helps in focusing the detection on the most manipulated part of deepfake videos, the face. The combination of CNN for feature extraction and LSTM for sequence processing forms a robust approach to deepfake detection.

Explanation:

Sequential Analysis: By examining frames in order, the model can detect unnatural transitions or alterations that might not be visible in a single frame.

Behavioural Patterns: Temporal features can capture subtle behavioural cues, like facial expressions or movements, that are difficult to fake consistently across multiple frames.

Temporal Coherence: Real videos have a natural flow of motion and expression, while deepfakes may exhibit temporal discrepancies due to the artificial generation of content.

By learning these temporal features, the system can more accurately classify a video as real or deepfake.

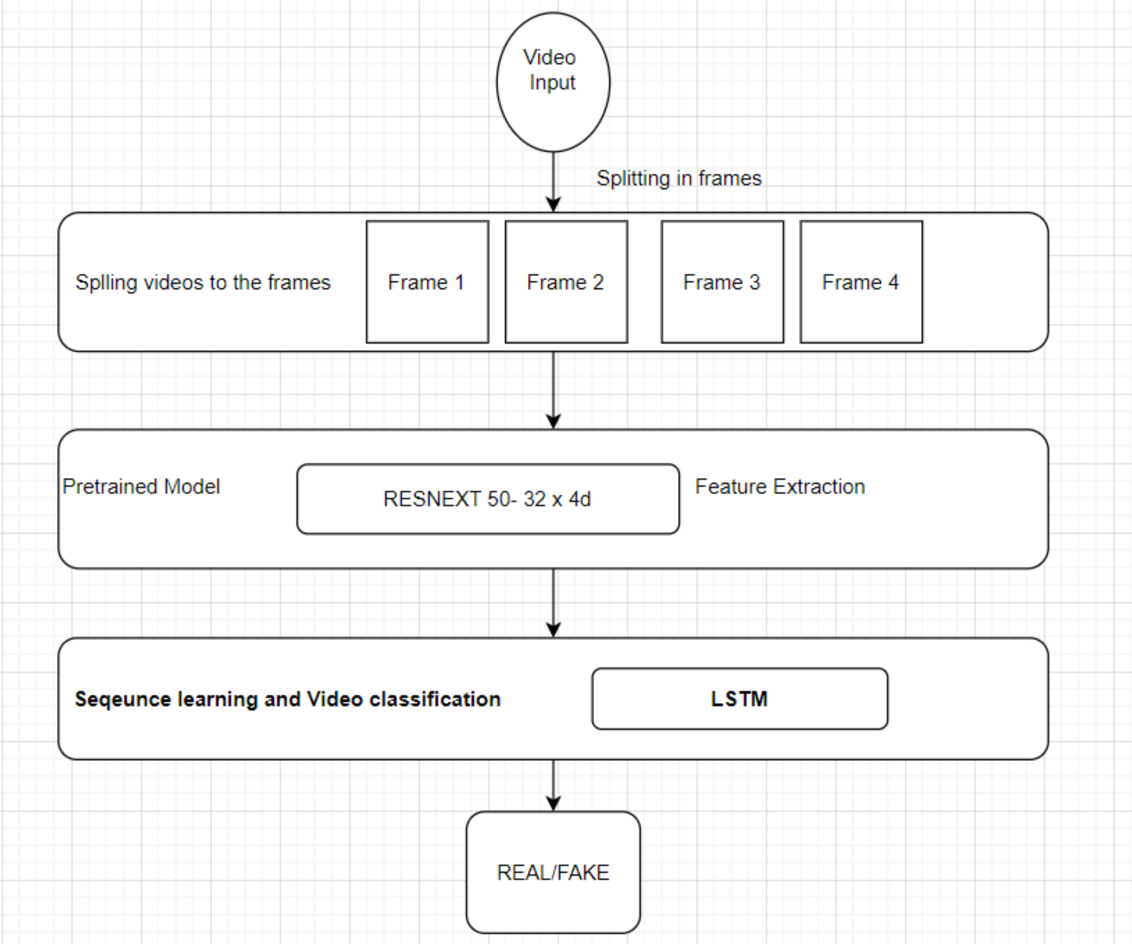

Image 1: Model Architecture

Results:

Model Name	Sequence	epochs	Train accuracy	Valid accuracy
LSTM model	20	20	99.17	86.15

Results achieved was very good as the train accuracy of model was quite extraordinary reaching 99.17 and valid accuracy peaked around north of 87 percent.

This shows our model is doing good the data that it has seen but not very good with the data it has not seen.

To further mitigate this problem we've to keep on using different parameters and see if we can reach the correct valid and test accuracy.

Reference: (jadhav, 2020): https://abhijithjadhav.medium.com/deepfake-video-detection-using-long-short-term-memory-df3674f83ecc.